

Generic 1 bit MUX

return smallest N such that

$$2^N \geq INS$$

input [$\lceil \log_2(INS) - 1 : 0 \rceil$]

ceiling
to call system function

Integer k ; outside course it's
behavioural & not
structural

```
always @ (——)
begin
  f = 'bx;
  for (k = ——)
    if (k == s)
      f = w[k];
end
```

so always @ → procedural aka
behavioural
↳ can't use gen. & has to use
integer

assign — = — → continuous aka
structural

↳ can use generic &

Conditional Control Constructs

① Routing Network

- Priority RW (if-else, non-parallel case)
- Multiplexing network (parallel case)

↳ In combinational

- No storage element

In an always element

- latch might be inferred
cause variables will keep it's
previous value if not assigned a
new value.

- cover all cases
(else in if & default in
case)